



**GETTING STARTED WITH
VERSION CONTROL:
INTRO TO GIT AND GITHUB**



**DATA
CULTURE
SOCIETY**

Support for **data-led** and **applied digital research** across the arts, humanities and social sciences.



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Volunteers from DH & RSE Summer School



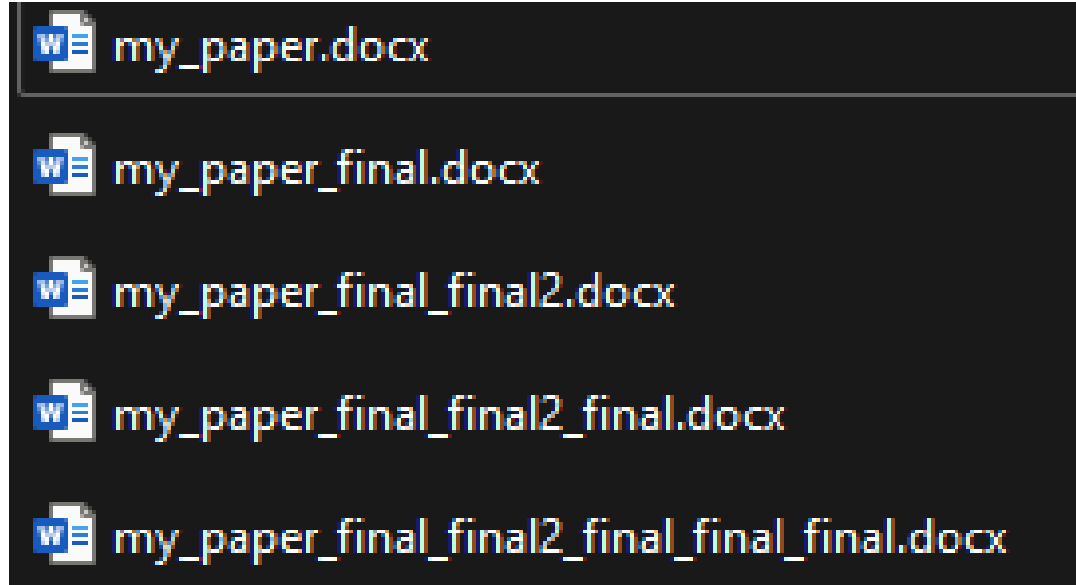
Plan for Today

- 09:10** Welcome & introduction
- 09:30** *Live Demo & troubleshooting:* Create your first repo & your first commit
- 10:00** *Live Demo & troubleshooting:* Create a new branch and merge it
- 10:30** Break
- 11:00** *Live Demo & troubleshooting:* Fork a repo & create a pull request
- 11:30** *Live Demo & troubleshooting:* Resolving a merge conflict
- 12:00** Generative AI and GitHub
- 12:10** Wrap up & additional resources
- 12:30** End of session / Lunch



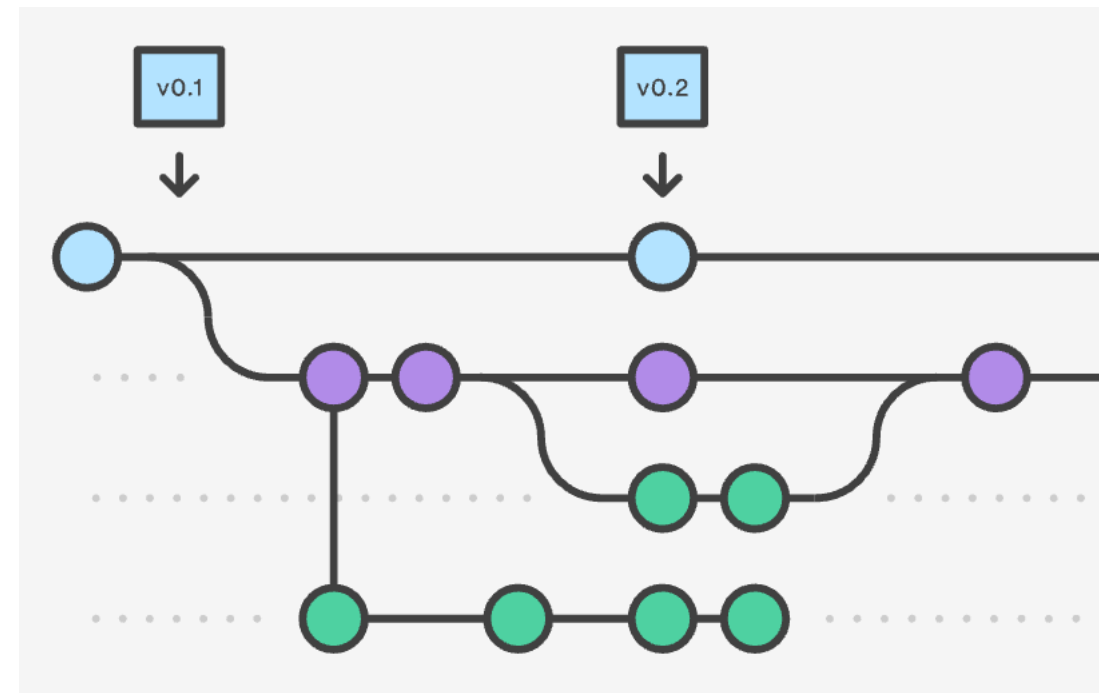
Version Control

- **Version control** helps you track changes to your files.
- You probably already have done (poor) version control before!



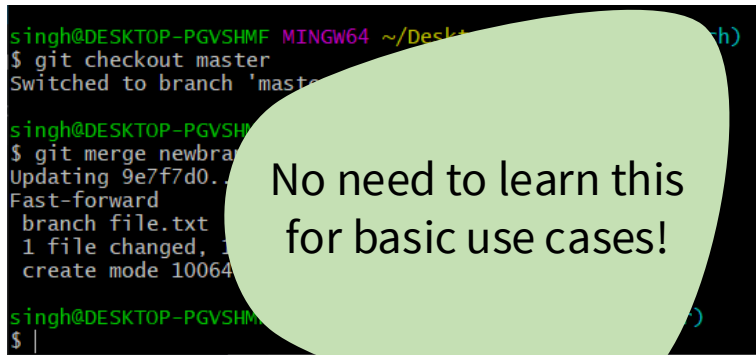
Version Control

- **Version control is the practice of tracking and managing *changes* to files** (traditionally, software code).
- Version control tracks every individual change by each contributor and helps prevent *concurrent* work from conflicting.
- It is a standard *workflow* in the tech industry.



git

Git is a version control system installed and maintained on your local system

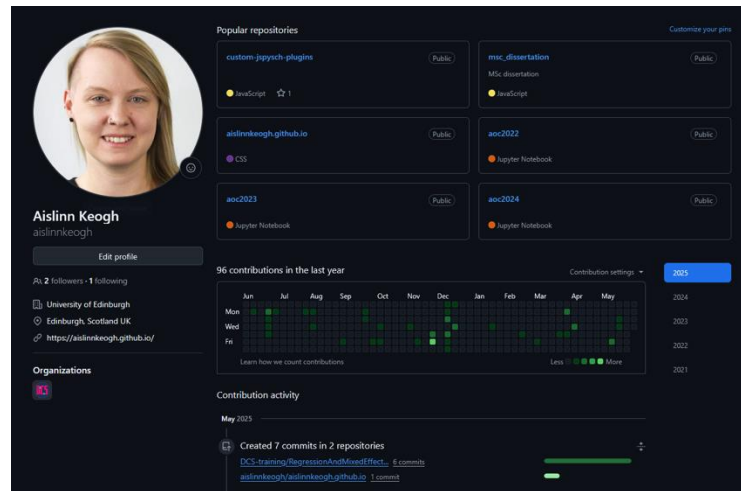


No need to learn this for basic use cases!

Image from <https://funkyvast.weebly.com/what-is-git-bash-terminal.html>

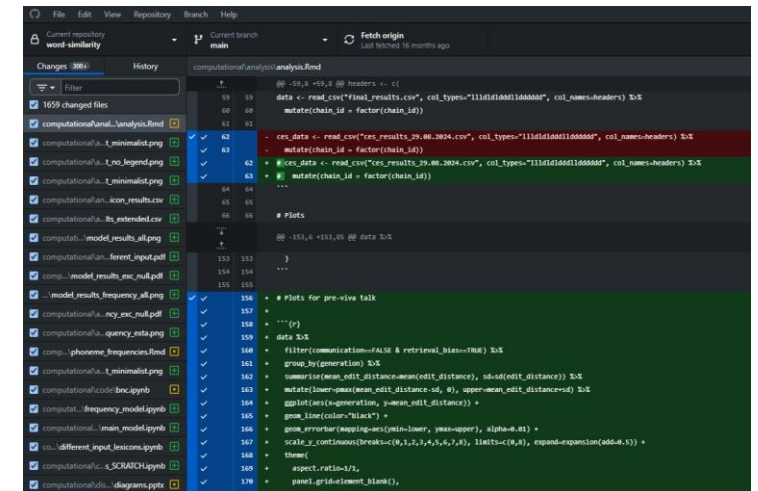
GitHub

GitHub is a hosting service for Git repositories. It is exclusively cloud-based



GitHub Desktop

A desktop application for working with git and GitHub on your computer



Terminology

repository/repo

commit

clone

branch

merge

fetch

pull

push

fork



Confused Person - ClipArt B...
clipartbest.com



Confused | Free Images at Clker...
clker.com



Confused Look Emoticon | ww...
imgkid.com



Confused Man Fre...
publicdomainpictur...



http://www.dreamstime.com/royalty-...
happyhormonesforlife.com



Confuse gesture, Confusion Medical s...
pngegg.com



Image Of Confused Person...
cliparts.co



Confused Cartoon ...
cliparts.co



Textusa: Perhaps... Confusing
textusa.blogspot.com



Confused person png imag...
lovepik.com



January | 2010 | Investing Caffeine
investingcaffeine.com



Principals Share The Most Bizarre ...
humaverse.com



Don't Worry, We're All Confused | Show...
showmeinstitute.org



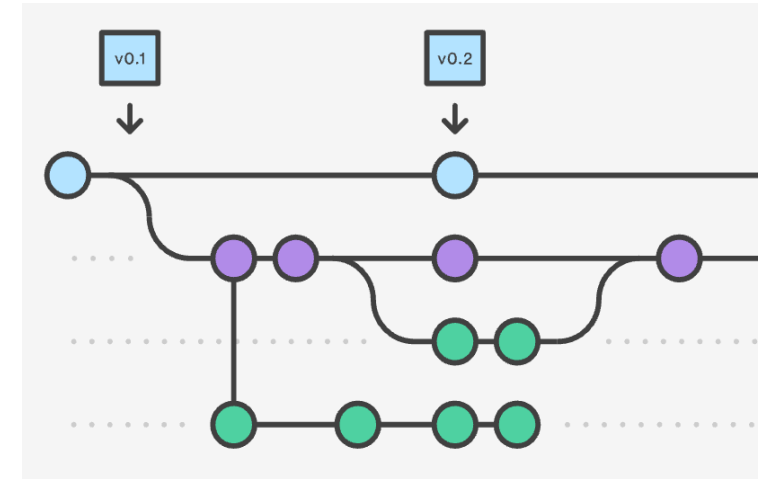
confused frustrated business man dreamst...
jobcrusher.com

Terminology

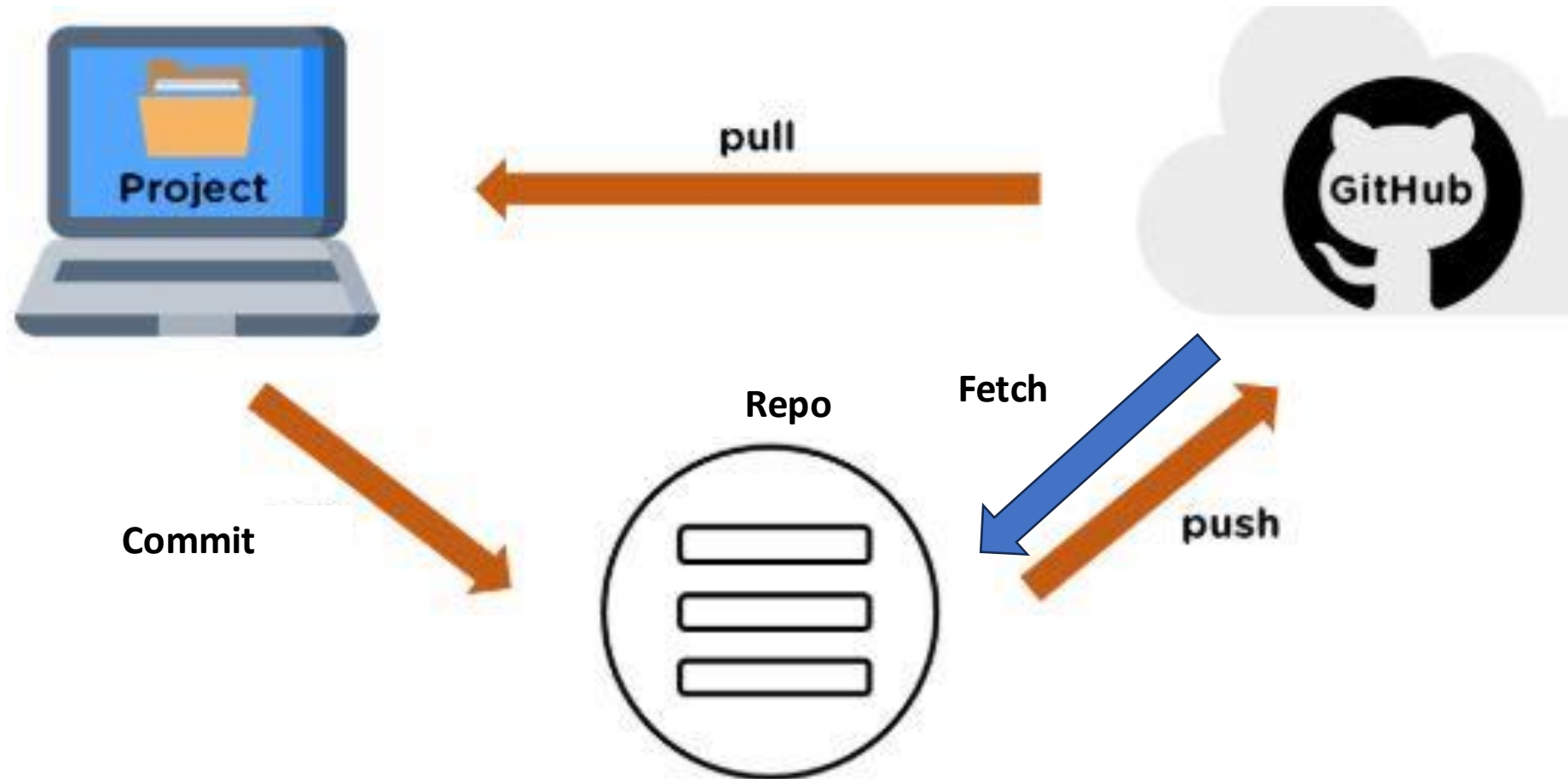
**repository/
repo** → a “folder” for your code for one project
It's a place where you can store your code,
your files, and each file's revision history.
Repositories can have multiple
collaborators and can be either public or
private.

clone → a copy of a repo (e.g. on your computer)

branch → a parallel version of a repo (repos can
have multiple branches)



Terminology



Viewing an existing repo

1. Go to www.github.com
2. Search for “London transport”
3. Add a Language filter for “Python”
4. Select “ **datasets/london-transport**” (updated 1 May or later)
<https://github.com/datasets/london-transport>
5. Describe what sort of files and folders it contains.



london-transport Public

Watch 6 Fork 5 Starred 5

main 4 Branches 0 Tags

Go to file

Add file

Code

Automated commit	Automated commit	2bf00c4 · 2 months ago	14 Commits
.github/workflows	Refresh london transport journeys and workflow	3 months ago	
data	Automated commit	2 months ago	
scripts	[update][s] Updating the script	last year	
README.md	[update][l] Updating the codebase	last year	
UPDATE_SCRIPT_MAINTENANCE_REPORT....	Refresh london transport journeys and workflow	3 months ago	
datapackage.json	Add top-level datapackage descriptor	3 months ago	

README

View on datahub.io

London public journeys by type of transport - this dataset was scrapped from [London data](#)

Data

The dataset is inside data folder. The data presents number of journeys on the public transport network by TFL reporting period, by type of transport. Data is broken down by bus, underground, DLR, tram, Overground and cable car.

- Period lengths are different in periods 1 and 13, and the data is not adjusted to account for that.
- Docklands Light Railway journeys are based on automatic passenger counts at stations.

About

No description, website, or topics provided.

Readme

Activity

Custom properties

5 stars

6 watching

5 forks

Report repository

Releases

No releases published

Packages

No packages published

Contributors 5



Languages

Python 100.0%

Cloning a repo with GitHub Desktop

1. Install GitHub Desktop from <https://desktop.github.com> if you haven't yet
2. Open GitHub Desktop, and log in with your account.
3. Locate the GitHub repository and click  then “Clone: Open with GitHub Desktop”.
4. Choose the location (local path) of your repo on your machine and proceed.
5. Describe what sort of files and folders it contains.



Creating your first repo

1. Go to www.github.com
2. Sign in / sign up
3. Click the + in the top right corner and select “New repository”
4. Choose a repository name, check “Add a README file”, and leave all other fields.
5. Click “Create repository”



Cloning your repo with GitHub Desktop

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Making a commit

1. Open the README.md file in a text editor of your choice
If you're unsure: on Windows use **Notepad**; on Mac use **TextEdit**
 - If you can't find the README.md file in your repo folder, you can create a README.md file using the editor.
2. Make some changes in your README file and save them
3. In GitHub Desktop, describe your commit and click commit when ready
4. Push to GitHub



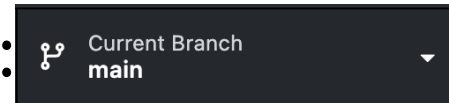
Terminology

repository/repo	→ a “folder” for your code for one project
clone	→ a copy of a repo (e.g. on your computer)
commit	→ saving your changes in the project
push	→ uploading your changes (your commits) to the server
fetch	→ downloading the latest changes from the remote repo
pull	→ integrating the latest changes into your clone of the repo
branch	→ a parallel version of a repo (repos can have multiple branches)
merge	→ combine the changes from one branch into another
fork	



Creating a new branch and merging it

1. In GitHub Desktop, open the branch menu:
2. Select 'new branch' and enter a name of your choice. Confirm that this branch is now displayed under 'Current Branch'.
3. Publish the new branch.
4. Make some changes in your README file. Save, commit, push.
5. Go to Branch -> Merge into the current branch and select your new branch. Create a merge commit, then push.



The background is a collage of three images. On the left is a circular inset showing a woman with dark hair and a pink flower in it, looking over a white object. In the center is a black and white photograph of several people in a room, looking at a large map or diagram on the wall. On the right is a black and white photograph of a person in a dark jacket holding a small object. The word 'Break' is centered over the collage in a large, bold, blue font.

Break

The background is a collage of three images. On the left is a circular inset showing a woman with dark hair and a pink flower in it, looking over a white object. In the center is a black and white photograph of several people in a room, looking at a large map or diagram on the wall. On the right is a black and white photograph of a person in a dark jacket holding a small object. The word 'Break' is centered over the collage in a large, bold, blue font.

**GETTING STARTED WITH
VERSION CONTROL: INTRO
INTRO TO GIT AND GITHUB**

Terminology

- repository/repo** → a “folder” for your code for one project
- clone** → a copy of a repo (e.g. on your computer)
- commit** → saving your changes in the project
- push** → uploading your changes (your commits) to the server
- fetch** → downloading the latest changes from the remote repo
- pull** → integrating the latest changes into your clone of the repo
- branch** → a parallel version of a repo (repos can have multiple branches)
- merge** → combine the changes from one branch into another
- fork** → a personal copy of someone else’s repo



Live Demo:

Forking a repo, making a commit, and making a pull request

1. Find a partner and get the link to their GitHub repo. Fork it on GitHub, then add your fork to GitHub Desktop.
2. Make a change to their README, commit, and push.
3. On GitHub, view your fork and open a pull request.
4. Review and approve each other's pull requests.



Live Demo:

Resolving a merge conflict

1. Choose your or your partner's repo.
2. In your fork of your partner's repo, make changes to README, commit, and push to your own fork.
3. Your partner also makes changes to the README, commit, and push to their own repo.
4. Make a pull request for your partner's repo. This will create a merge conflict. Resolve as demonstrated.



Generative AI and GitHub



There are many ways that people use **GitHub Copilot**, including:

1. Get **code suggestions** as you type
2. **Chat** with Copilot to get help (browser/command line)
3. Generate **descriptions of changes** in a pull request
4. Create pull requests for you to **review** (and vice versa)

Alternatives: Claude Code, Cursor and [others](#)



Generative AI and GitHub



Why use it?	How to be careful	Why not use it?
• Productivity boost • Integration • Learning tool • Built-in security	• Validate every line • Turn off auto complete • Data privacy • Check local rules	• Subtle errors • Project too complex • Costs • Cloud dependency



Additional Resources

- GitHub online tutorials
 - Introduction (tutorial repeating some of what we did today): <https://github.com/skills/introduction-to-github>
 - Reviewing pull requests: <https://github.com/skills/review-pull-requests>
 - Resolving merge conflicts: <https://github.com/skills/resolve-merge-conflicts>
 - List of all skills tutorials: <https://github.com/skills>
- GitHub student pack: <https://education.github.com/pack>

